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Class : CSE

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Batch : 2025-29

Assignment : 14

Problem Statement : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Code and Output Screenshots

```
1 public class TryCatchExample {
2
3     public static void main (String[] args) {
4         int num1 = 67;
5         int num2 = 0;
6
7         try {
8             int result = num1 / num2;
9
10            System.out.println("Result:" + result);
11        }
12        catch (ArithmeticException e) {
13
14            System.out.println("Error: Cannot divide by zero!");
15            System.out.println("Exception message: " + e.getMessage());
16        }
17
18        System.out.println("Program continues after exception handling.");
19    }
20 }
```

OUTPUT

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac TryCatchExample.java && java TryCatchExample
Error: Cannot divide by zero!
Exception message: / by zero
Program continues after exception handling.
[Done] exited with code=0 in 0.657 seconds
```



```
MultipleExceptions.java > ...
1 public class MultipleExceptions {
    Run | Debug | Run main | Debug main
2     public static void main(String[] args) {
3         try {
4             String str = "6767abc";
5             int num = Integer.parseInt(str);
6
7             int res = 100/0;
8         } catch (NumberFormatException | ArithmeticException e) {
9             System.out.println("Error : " + e.getClass().getSimpleName() + " occurred.");
10        }
11        System.out.println("Program continues after exception handling.");
12    }
13 }
14
```

PROBLEMS 32 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac MultipleExceptions.java && java MultipleExceptions
Error : NumberFormatException occurred.
Program continues after exception handling.
[Done] exited with code=0 in 0.604 seconds
```

Conclusion

This program demonstrated how Java handles runtime errors using try and catch blocks. It showed the use of simple, multiple, and nested try-catch blocks to manage different types of exceptions effectively. Proper exception handling prevents abrupt program termination and makes the code more reliable and user-friendly. Overall, it improves program stability and helps in debugging and maintaining applications.



Exercises

Q1. Create an ATM program handling invalid withdrawal amount using try-catch.

Ans) Case 1 :

```
J ATM.java > Java > ATM
1  import java.util.Scanner;
2  public class ATM {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          int balance = 5000;
6          try {
7              System.out.print("Enter withdrawal amount: ");
8              int amount = sc.nextInt();
9              if (amount <= 0) {
10                 throw new Exception("Withdrawal amount must be greater than 0.");
11             }
12             if (amount > balance) {
13                 throw new Exception("Insufficient balance.");
14             }
15             balance -= amount;
16             System.out.println("Withdrawal Successful!");
17             System.out.println("Remaining Balance: " + balance);
18         }
19         catch (Exception e) {
20             System.out.println("Error: " + e.getMessage());
21         }
22         sc.close();
23     }
24 }
```

PROBLEMS 37 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac
● ATM.java } ; if ($?) { java ATM }
Enter withdrawal amount: 1400
Withdrawal Successful!
Remaining Balance: 3600
○ PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



Case 2

```
J ATM.java > Java > ATM
1  import java.util.Scanner;
2  public class ATM {
    Run | Debug | Run main | Debug main
3  public static void main(String[] args) {
4      Scanner sc = new Scanner(System.in);
5      int balance = 5000;
6      try {
7          System.out.print("Enter withdrawal amount: ");
8          int amount = sc.nextInt();
9          if (amount <= 0) {
10             throw new Exception("Withdrawal amount must be greater than 0.");
11         }
12         if (amount > balance) {
13             throw new Exception("Insufficient balance.");
14         }
15         balance -= amount;
16         System.out.println("Withdrawal Successful!");
17         System.out.println("Remaining Balance: " + balance);
18     }
19     catch (Exception e) {
20         System.out.println("Error: " + e.getMessage());
21     }
22     sc.close();
23 }
24 }
```

PROBLEMS 37 OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if (\$?) { javac AT
- Enter withdrawal amount: xyz
- Error: null
- PS C:\Users\LENOVO\Documents\JAVA Assignments>



Q2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Ans) Case 1

```
J Cart.java > Java > Cart
1  import java.util.Scanner;
2
3  public class Cart {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          try {
9              System.out.print("Enter product quantity: ");
10             int quantity = sc.nextInt();
11
12             if (quantity <= 0) {
13                 throw new Exception("Invalid product quantity.");
14             }
15
16             System.out.println("Order placed successfully.");
17         }
18         catch (Exception e) {
19             System.out.println(e.getMessage());
20         }
21
22         sc.close();
23     }
24 }
```

PROBLEMS 39 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code +

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Cart.java } ;
Enter product quantity: 15
Order placed successfully.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



Case 2

```
J Cart.java > Java >  Cart
1  import java.util.Scanner;
2
3  public class Cart {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          try {
9              System.out.print("Enter product quantity: ");
10             int quantity = sc.nextInt();
11
12             if (quantity <= 0) {
13                 throw new Exception("Invalid product quantity.");
14             }
15
16             System.out.println("Order placed successfully.");
17         }
18         catch (Exception e) {
19             System.out.println(e.getMessage());
20         }
21
22         sc.close();
23     }
24 }
```

PROBLEMS 39 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Cart.java }

Enter product quantity: -3
Invalid product quantity.
○ PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



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